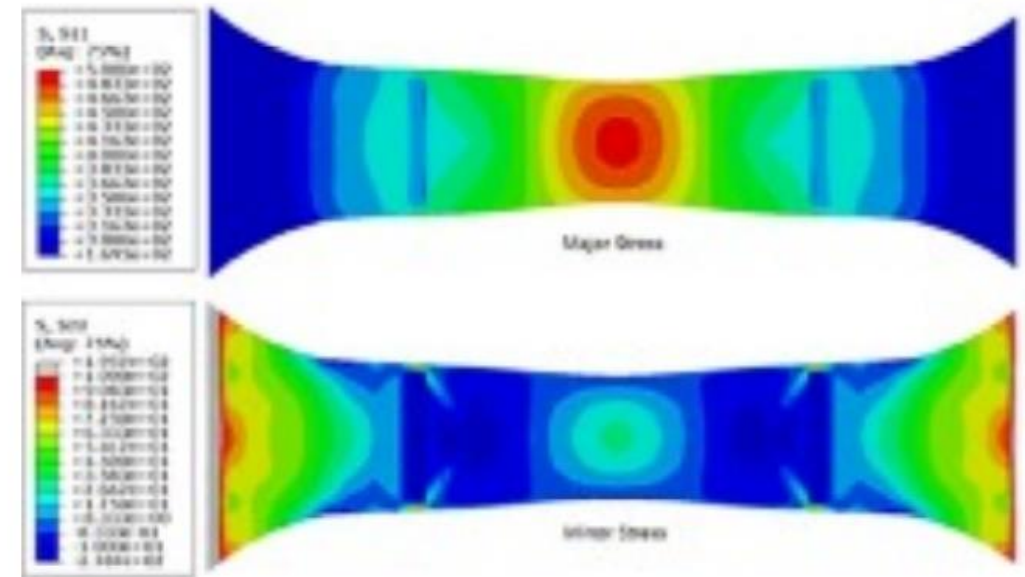


Materials Science 1 – Mechanical properties

Materials Science 1

Mechanical properties of materials

Review of previous knowledge



Learning objectives and Learning outcomes

- Review concepts of stress and strain
- Estimate mechanical properties from stress-strain diagrams
- Identify loading modes of everyday products
- Use mechanical properties diagrams to select materials

Poll – Answers in We Chat

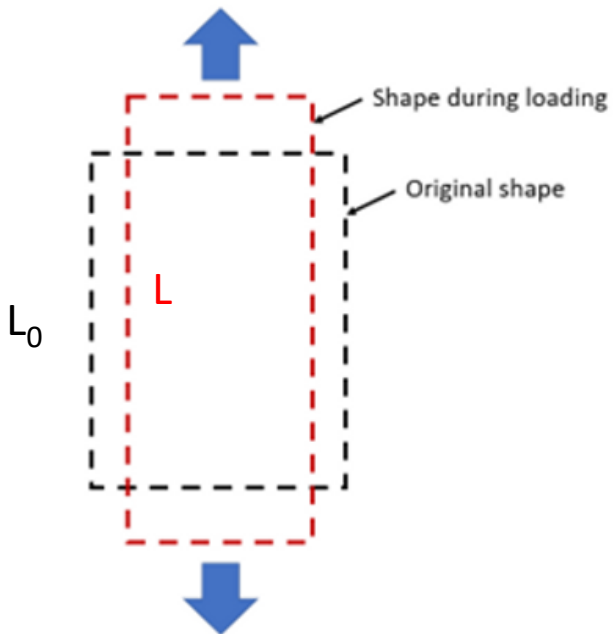
Q1) What is engineering strain?

- A) The relationship between change in length (initial vs final) of a material when a load is applied in a specific direction;
- B) The relationship between instantaneous load and a material cross sectional area.

Poll – Answer

Q1) What is engineering strain?

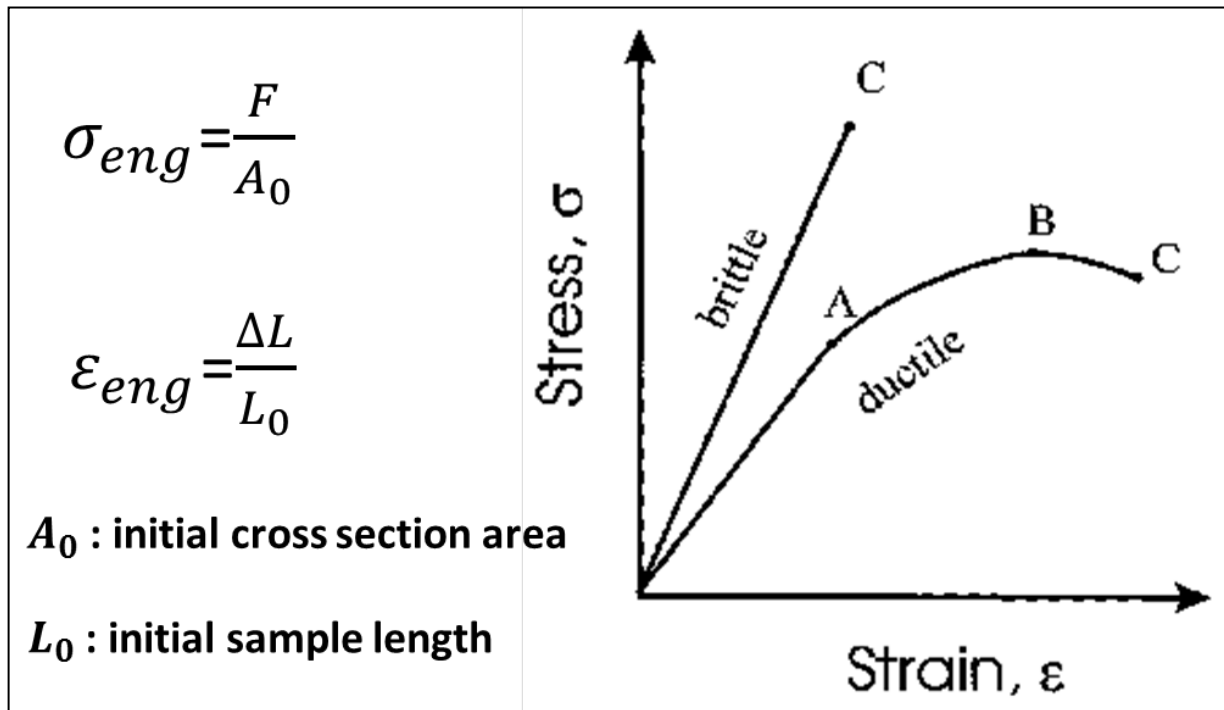
A) The relationship between change in length (initial vs final) of a material when a load is applied in a specific direction;



$$\epsilon_{eng} = \frac{\Delta L}{L_0} = \frac{L - L_0}{L_0}$$

Tensile test and stress-strain diagrams

- Samples with special shapes (normally dog-bone-like shapes) are pulled along their length and the force F applied and the elongation of the sample ($\Delta L = \text{current length} - \text{initial length}$) are simultaneously measured by the testing equipment (Instron machines are most common).
- The force-elongation ($F-\Delta L$) diagram can be converted in engineering stress-strain ($\sigma-\epsilon$) plot:



From the stress-strain diagram it is possible to distinguish two main categories of behaviour:

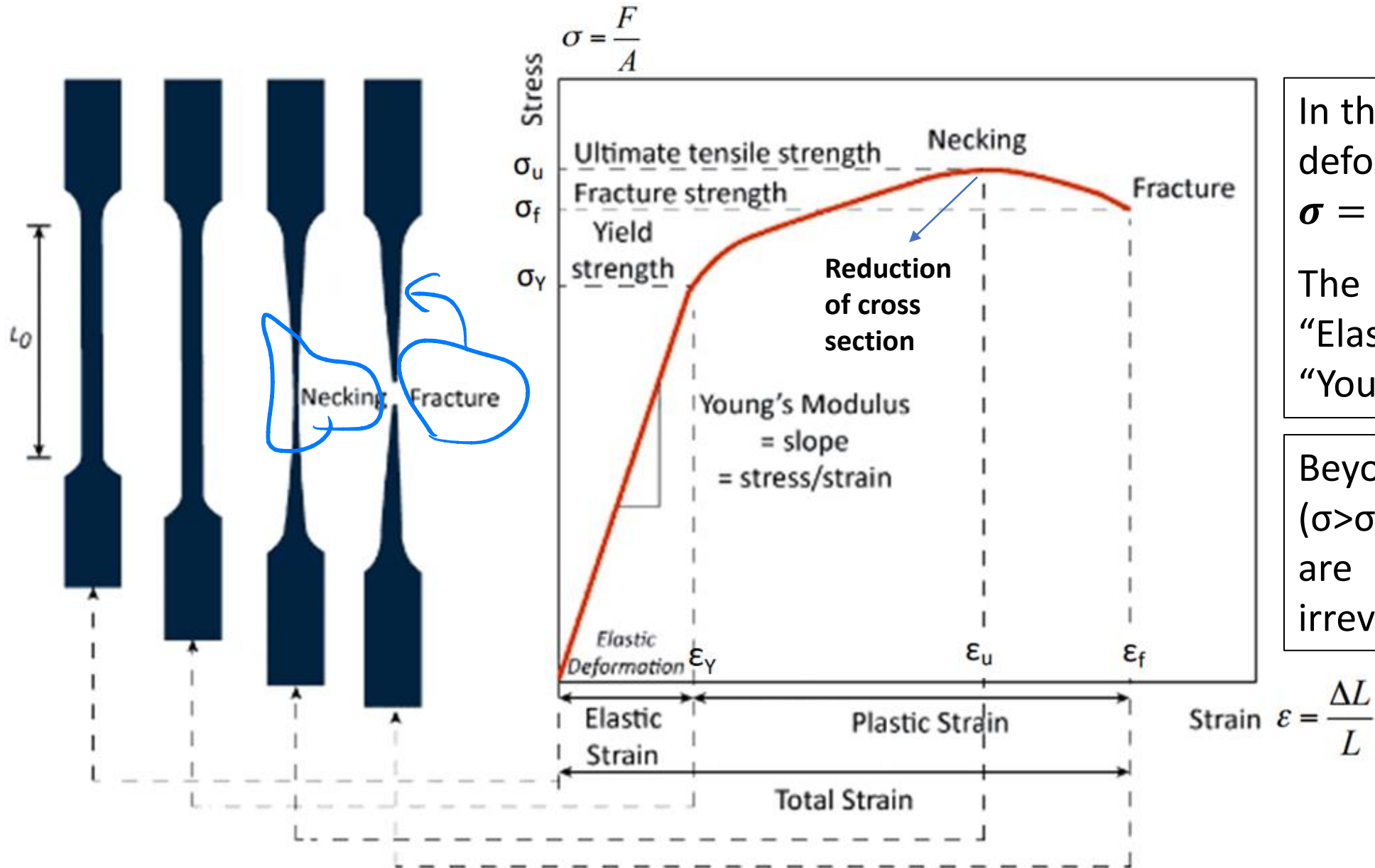
1. Brittle:

- Linear behaviour until failure;
- Reversible deformations (linear elastic).

2. Ductile:

- Non-linear behaviour beyond a certain point;
- Non-recoverable deformations (plastic).

Stress-strain diagram of ductile materials

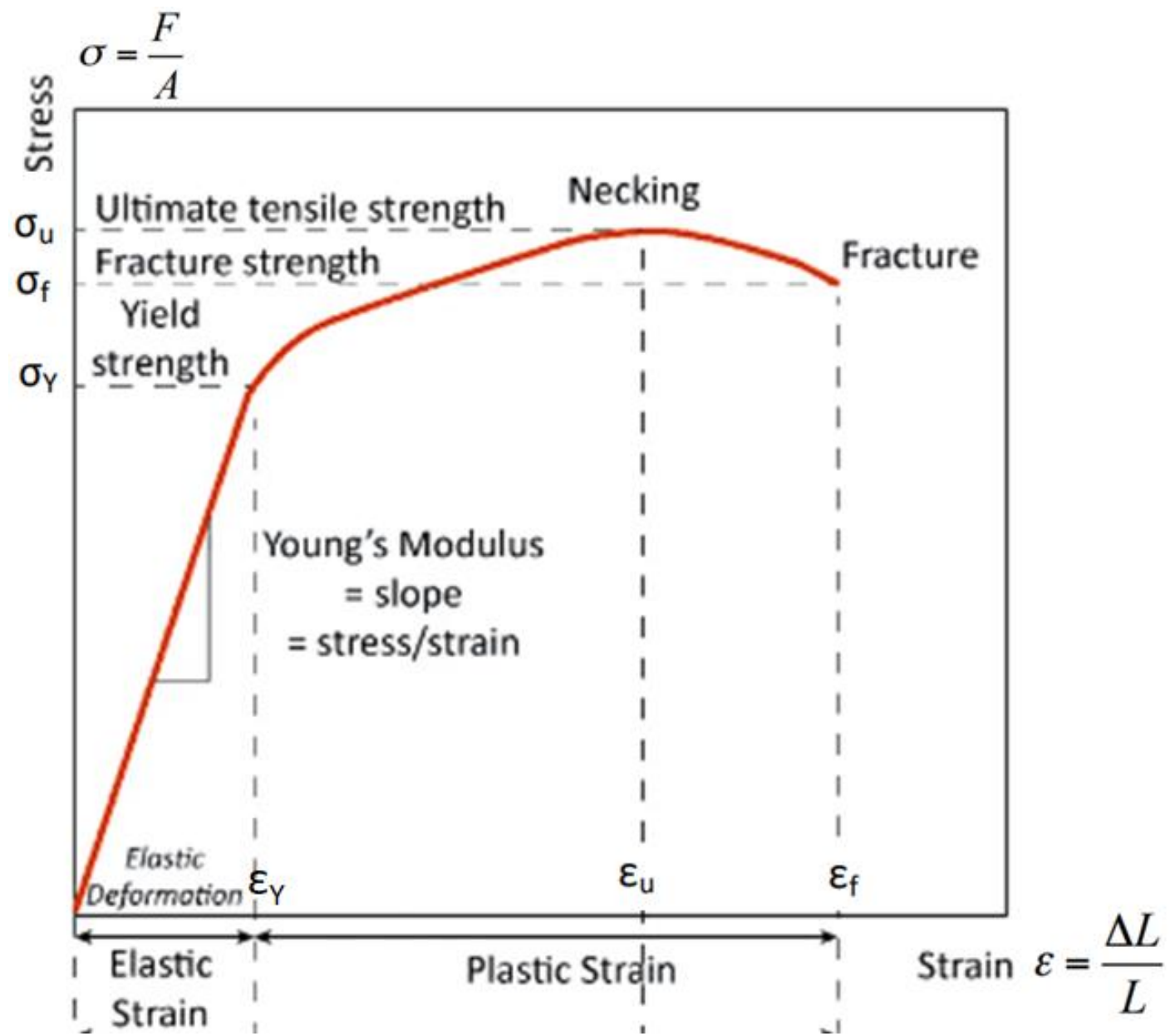


In the linear part, only elastic deformation take place:
 $\sigma = E\epsilon$ (Hooke's law).

The slope E is defined as "Elastic modulus" or "Young's modulus".

Beyond the linear part ($\sigma > \sigma_Y$), plastic deformations are produced, which are irreversible.

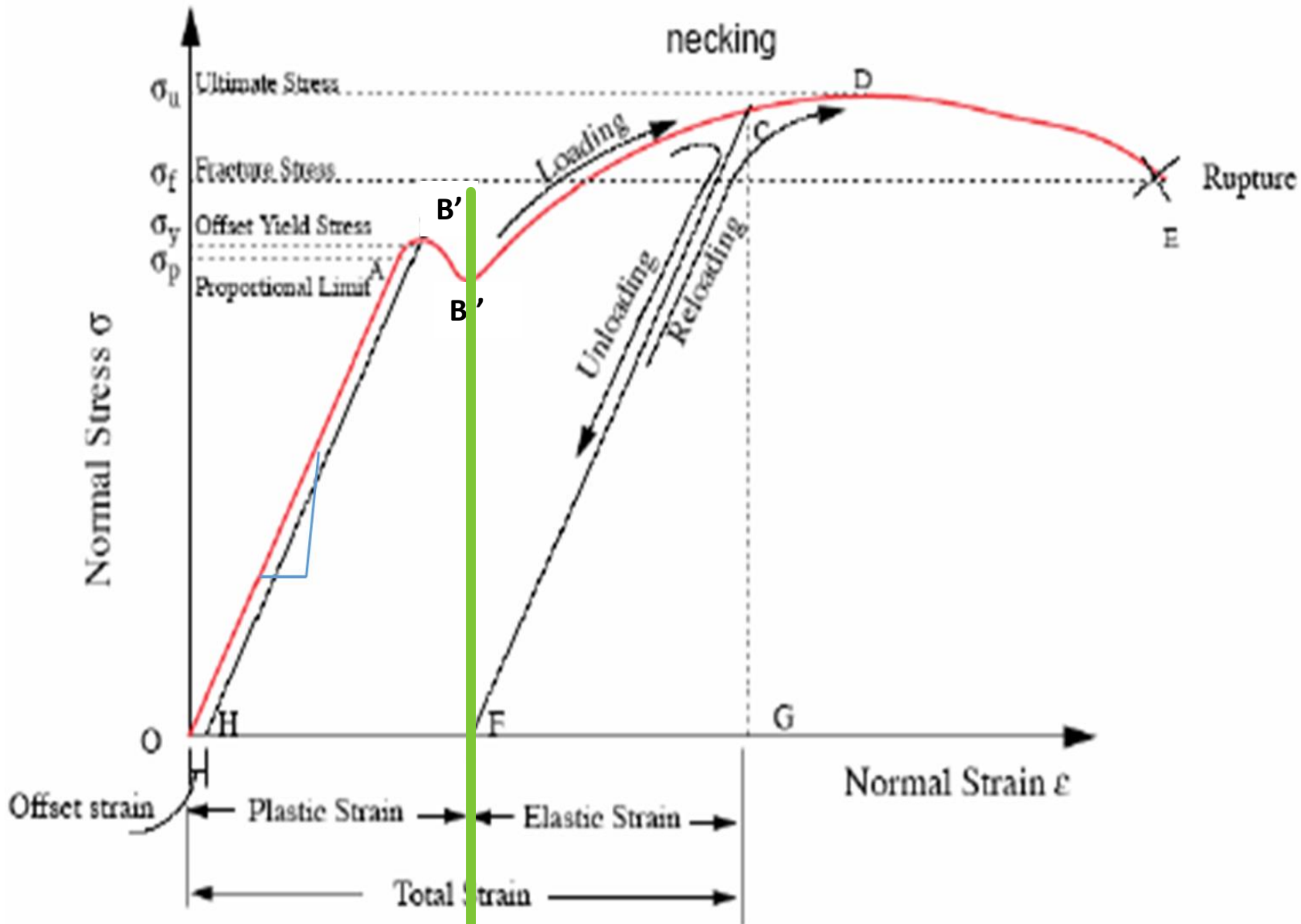
Stress-strain diagram of ductile materials



Important parameters

Symbol	Parameter
E	Elastic modulus (Young's modulus)
σ_y	Yield stress (elastic limit)
σ_f	Stress at failure
σ_u	Ultimate strength
ϵ_y	Yield strain (elastic limit)
ϵ_f	Strain at failure

Stress-strain diagram of ductile materials

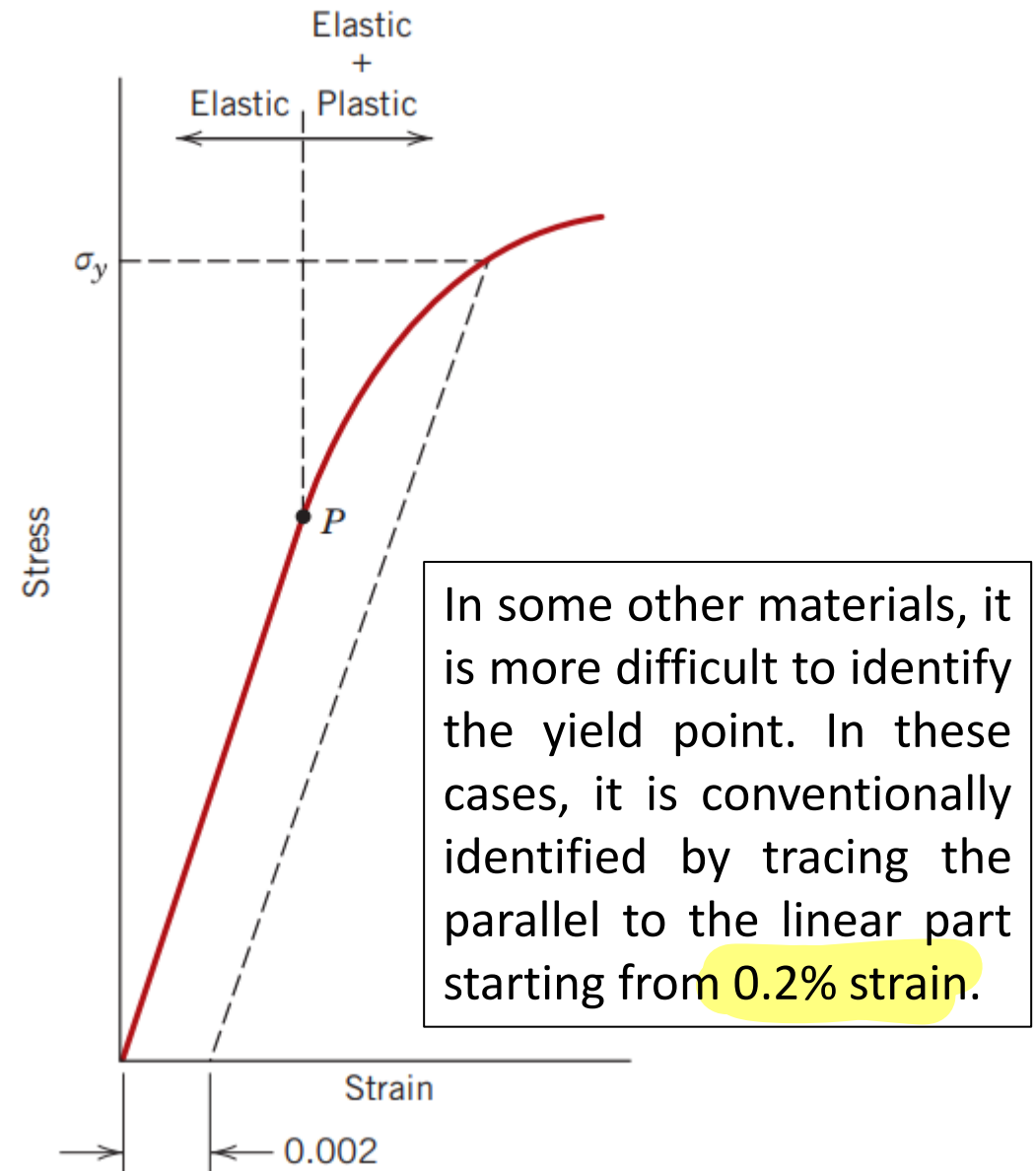
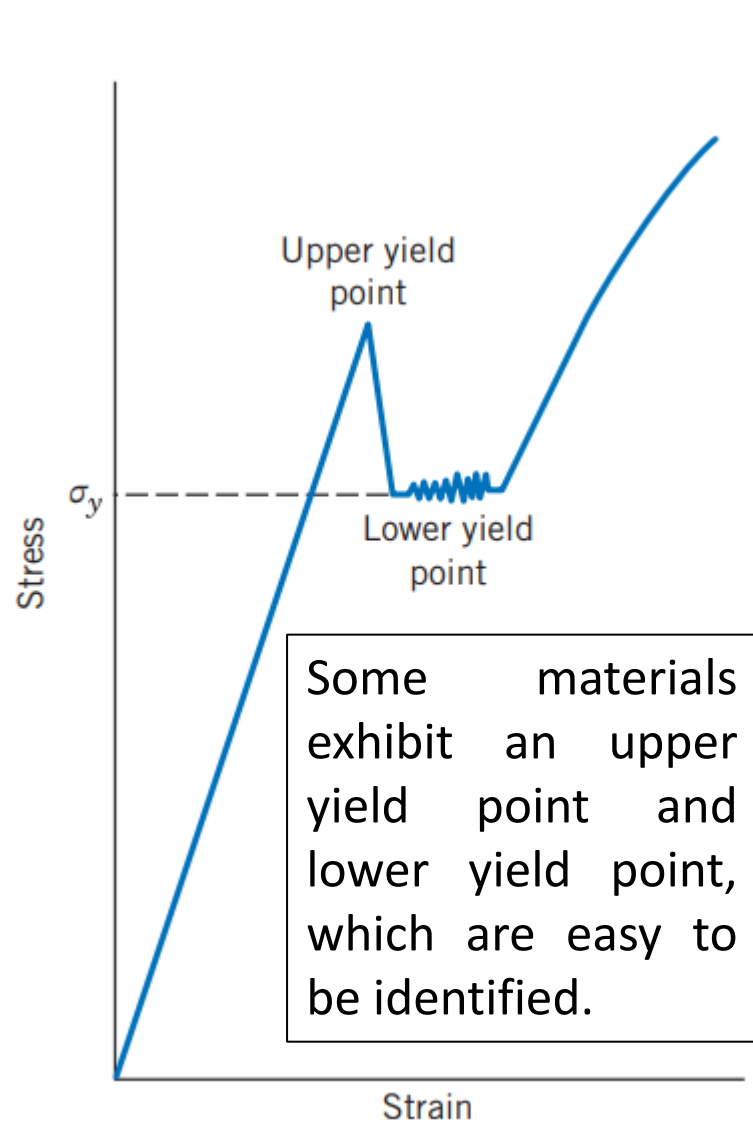


Some ductile materials show two yield points, defined upper (B') and lower yield point (B'').

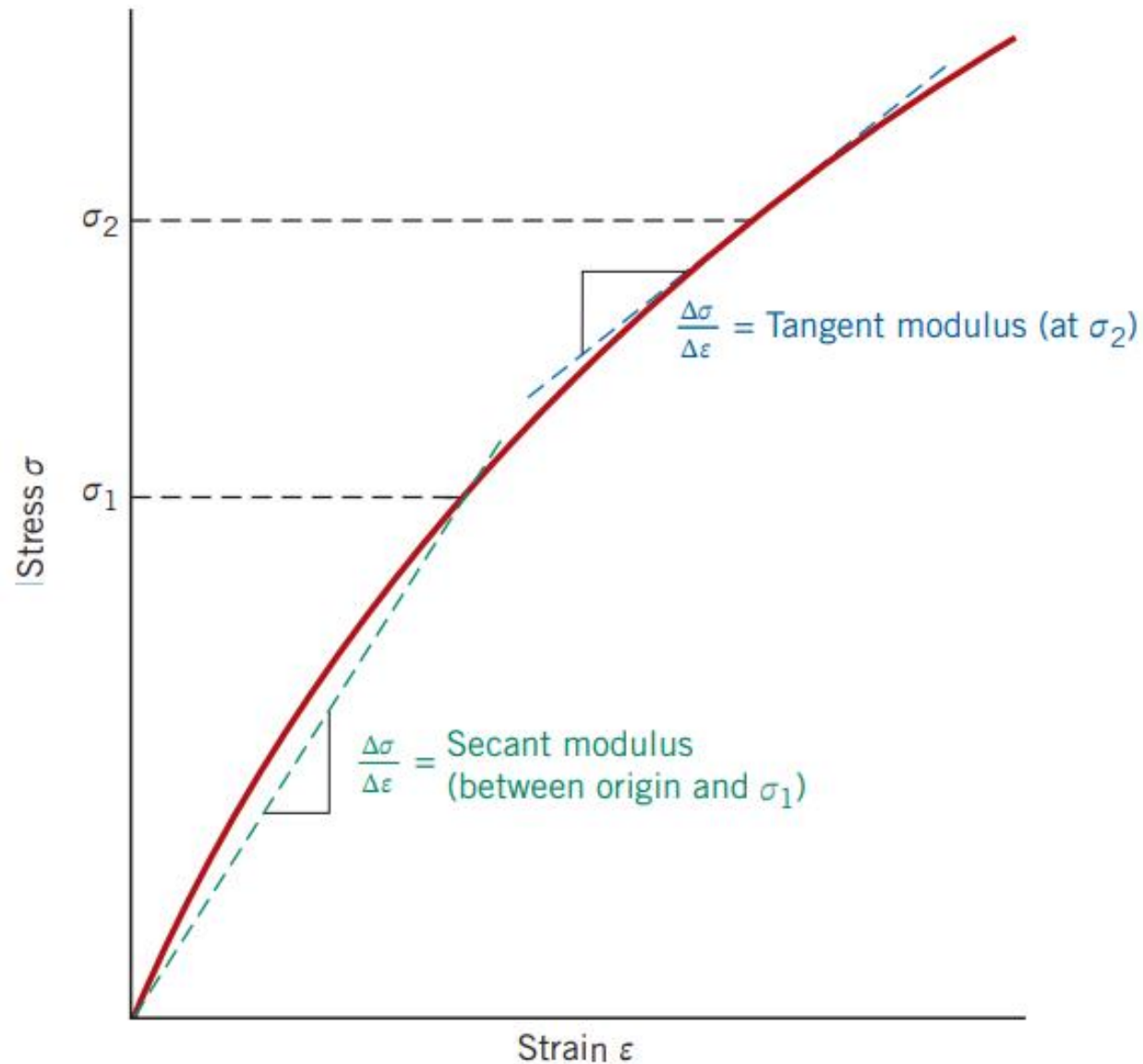
Necking: strain hardening does not compensate anymore the reduction of cross section, so stress starts to decrease.

If the load is removed from a point beyond the yield point, the elastic strain is recovered (segment FG), while the plastic strain is permanent (segment HF).

Stress-strain diagram of ductile materials



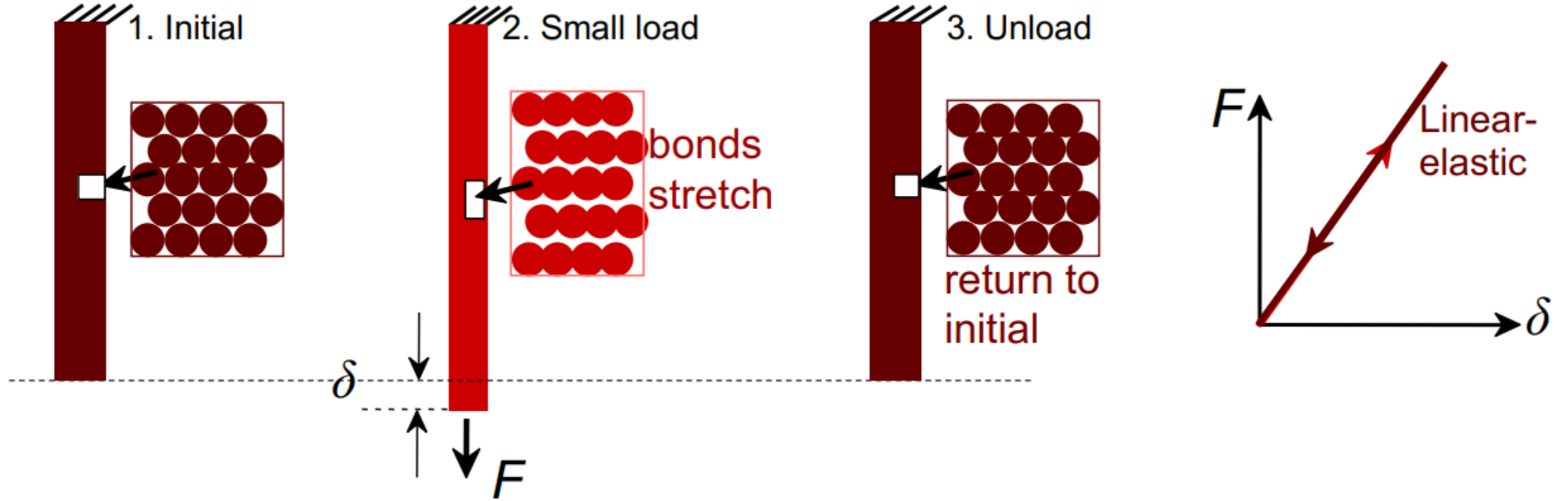
Stress-strain diagram of ductile materials



In some materials (i.e., gray cast iron, concrete, and many polymers) the elastic portion of the stress-strain curve is not linear; hence, it is not possible to determine a modulus of elasticity as described previously. For this nonlinear behaviour, either the **tangent or secant modulus** is normally used.

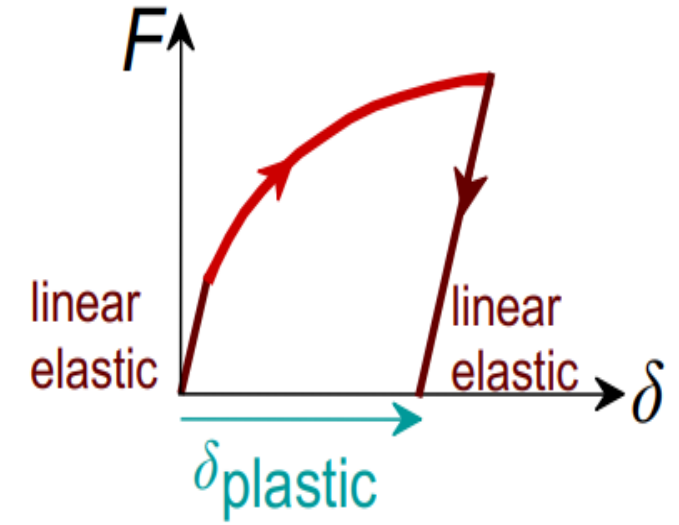
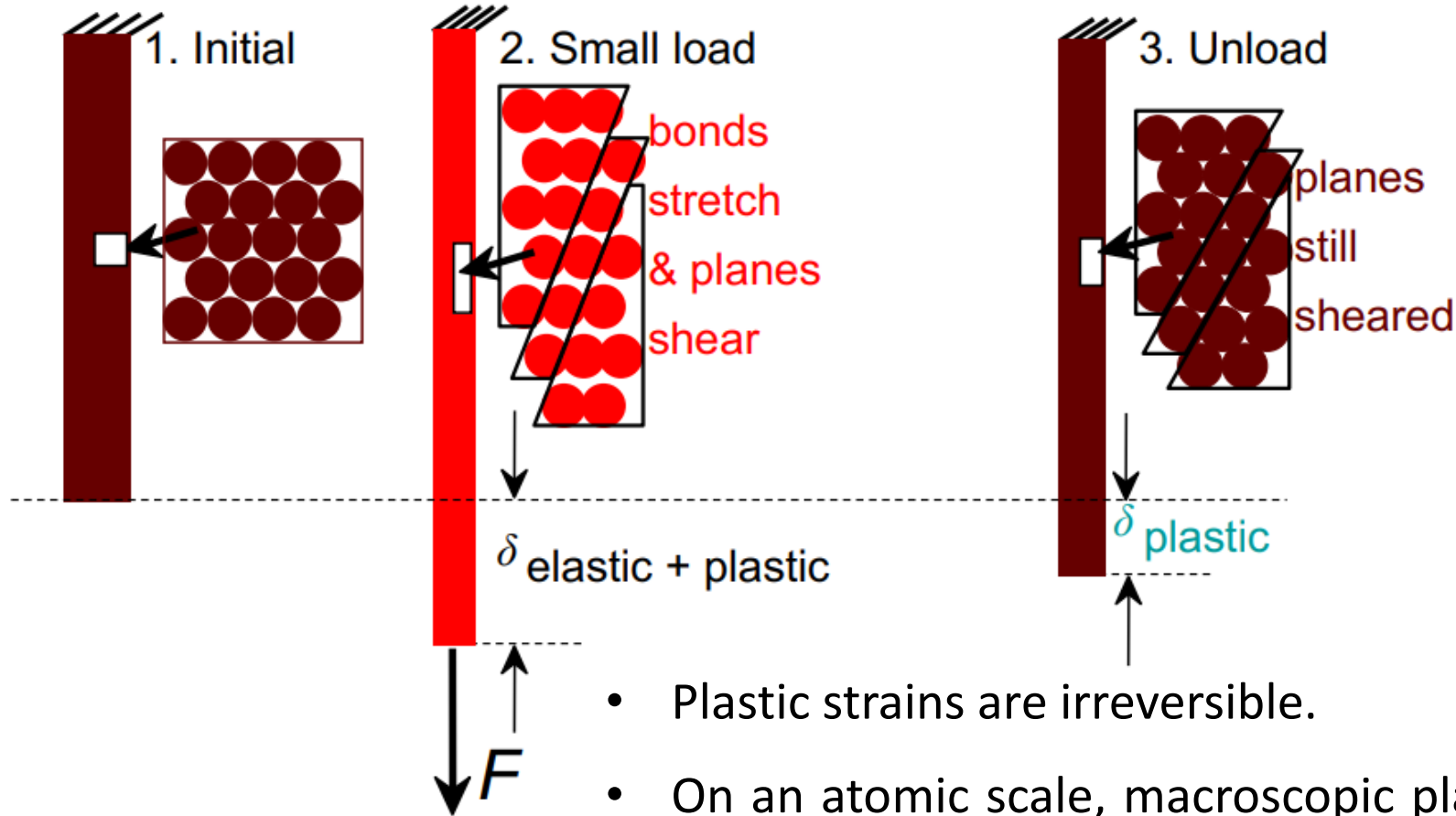
For these materials, also the strain offset method is not applicable, and the usual practice is to define the yield strength as the stress required to produce some amount of strain (e.g., $\epsilon = 0.005$).

Atomistic mechanisms of elastic deformations



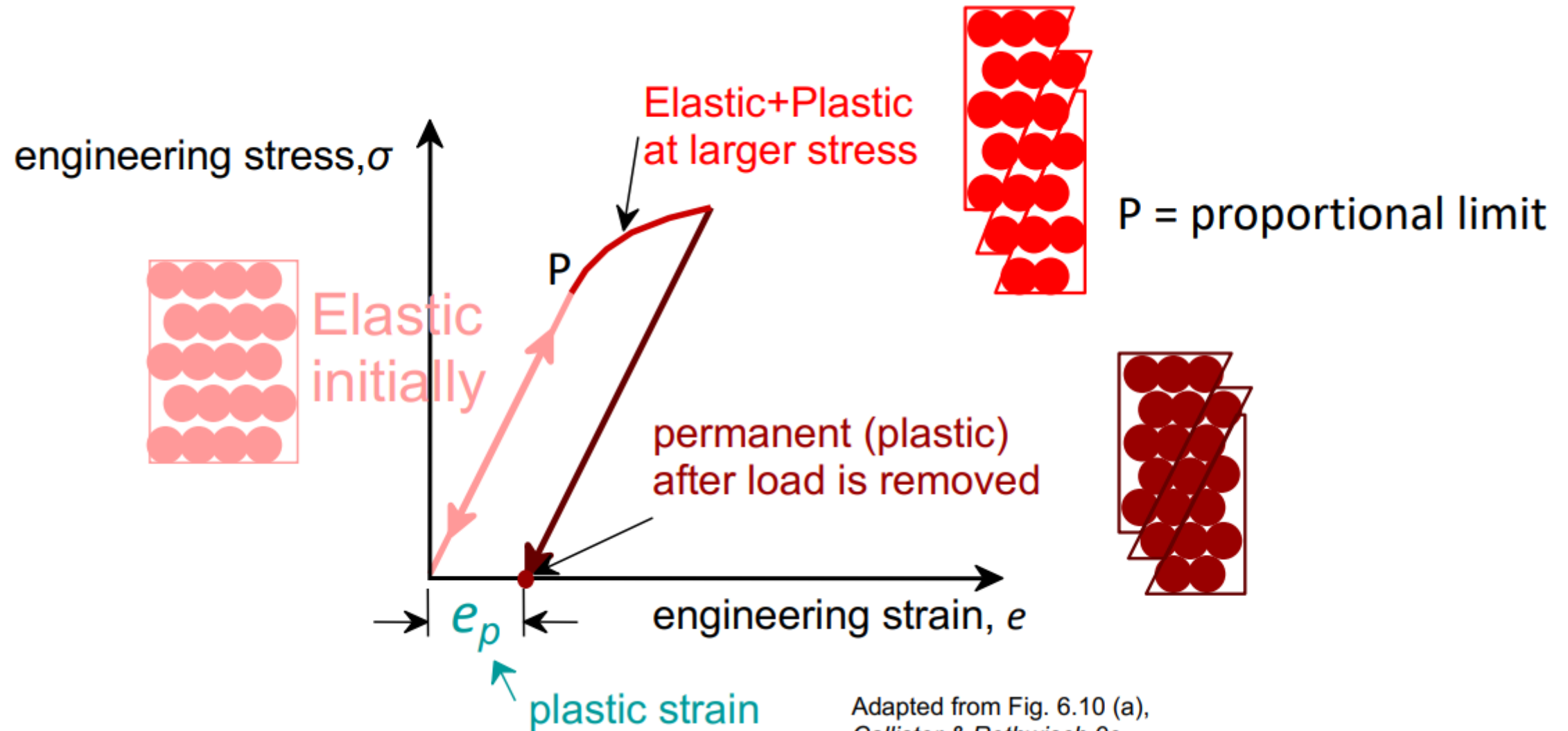
- Elastic strains are reversible.
- On an atomic scale, macroscopic elastic strains are due to small changes in the interatomic spacing and the reversible stretching of interatomic bonds.
- As a consequence, the modulus of elasticity is a measure of the resistance to separation of adjacent atoms.

Atomistic mechanisms of plastic deformations



- Plastic strains are irreversible.
- On an atomic scale, macroscopic plastic deformations correspond to the breaking of bonds between atoms originally neighbours and to the formation of new bonds with other neighbours.
- In crystalline solids, plastic deformation occurs via a process called slip, which involves the motion of dislocations, while in non-crystalline solids it occurs by a viscous flow mechanism.

Atomistic mechanisms of deformations



Poll – Answers in We Chat

Q1) Which family of materials show larger values of elastic deformation?

- A) Metals
- B) Ceramics
- C) Polymers

Poll – Answer

Q1) Which family of materials show larger values of elastic deformation?

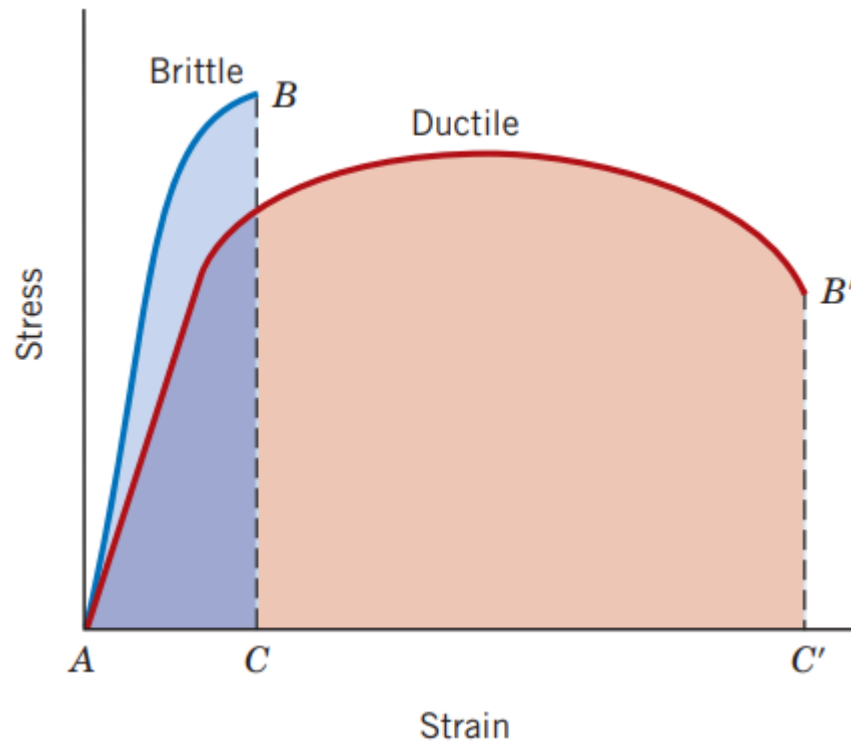
C) Polymers.

Polymers have long and flexible chains that can easily slide past each other when subjected to stress, allowing for significant elongation. The intermolecular forces in polymers, such as van der Waals forces and dipole-dipole interactions, are generally weaker compared to the strong metallic or ionic bonds found in metals and ceramics, making it easier for polymer chains to separate and slide past one another. Some polymers can be cross-linked. Cross-links are covalent bonds that connect polymer chains, providing additional strength and allowing the material to recover its shape after deformation.

Stress-strain diagram of ductile materials

Ductility

It is a measure of the degree of plastic deformation that the material has experienced until fracture. It is expressed quantitatively as either percentage of elongation or percentage of reduction in the cross-section area.



$$\%EL = \left(\frac{l_f - l_0}{l_0} \right) \times 100$$

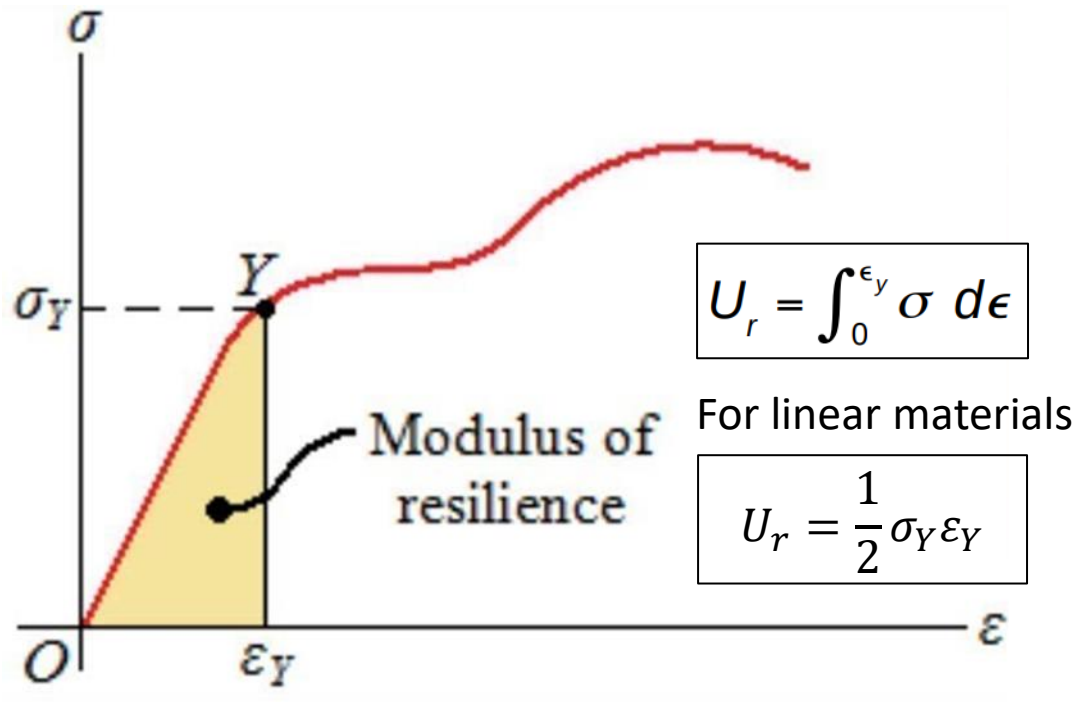
l_f =final length;
 l_0 =initial length.

$$\%RA = \left(\frac{A_0 - A_f}{A_0} \right) \times 100$$

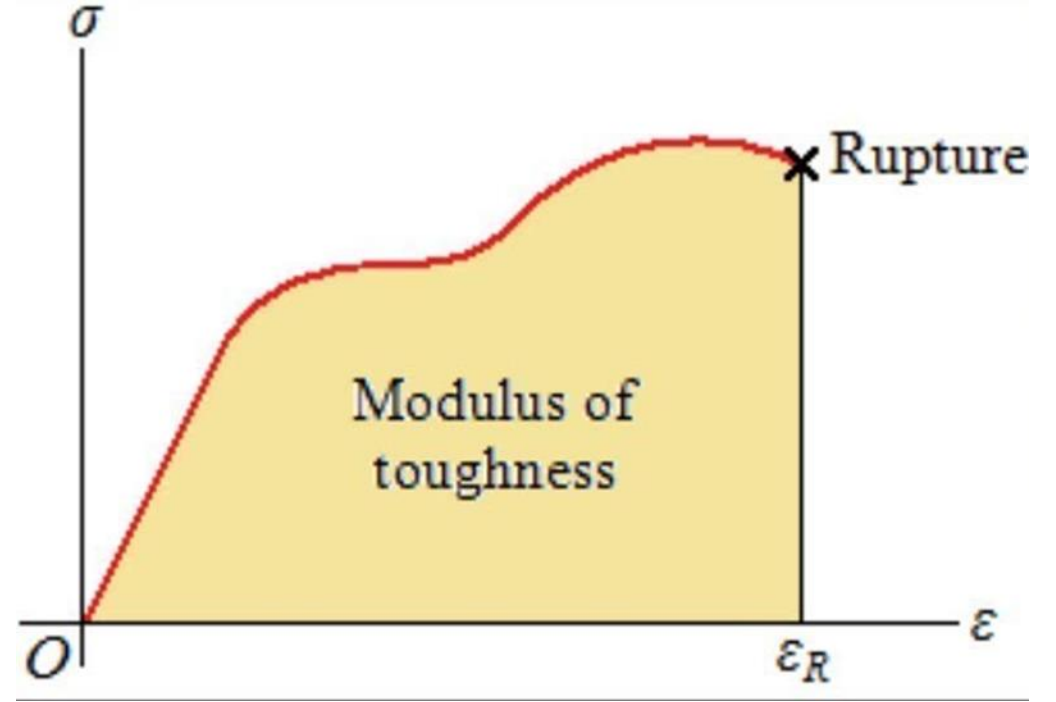
A_f =final cross section;
 A_0 =initial cross section.

Stress-strain diagram of ductile materials

- Modulus of resilience
- Modulus of toughness



Ability of material to absorb energy in the elastic region.



Ability of material to absorb energy until fracture.

Summary

- High elastic modulus – Rigid materials;
- High ultimate strength - Strong materials;
- Non-linear stress-strain diagram, high strain at failure and large shape variations (i.e. necking) – Ductile materials;
- Large area beneath the stress-strain diagram – Tough materials.
- Large area beneath the elastic region of the stress-strain diagram – Resilient materials.